Putnam Model in Software Engineering

Software Cost Estimation and Life Cycle Modeling

Overview

- • Proposed by Lawrence H. Putnam in 1978
- • Based on Rayleigh manpower distribution curve
- • Used for estimating effort, schedule, and cost
- • Implemented in SLIM (Software Life Cycle Management) tool

Core Concept

- • Effort follows a Rayleigh distribution:
- $E(t) = (K / td^3) * t^2 * e^{(-t^2 / td^2)}$

- • Effort starts low, rises to a peak, then declines
- • Represents real manpower distribution in a project

Software Equation

- $S = Ck \times (E)^{(1/3)} \times (td)^{(4/3)}$

- Where:
- • S = Software size (LOC or function points)
- • E = Total effort (person-years)
- • td = Development time (years)
- • Ck = Technology constant

- Effort: $E = (S / (Ck \times td^{(4/3)}))^3$

Interpretation and Key Parameters

- • Decreasing schedule increases effort rapidly
- • Realistic schedules reduce total effort

- Parameters:
- S - Software size
- td - Development time
- E - Total effort
- Ck - Technology constant

Advantages and Limitations

- Advantages:
- • Realistic manpower modeling
- • Highlights time–effort trade-offs
- • Useful for project control

- Limitations:
- • Needs historical calibration
- • Not suitable for small or changing projects
- • Assumes stable scope

Example

- Given:
- S = 100,000 LOC, Ck = 2000, td = 2 years

- $E = (100{,}000 / (2000 \times 2^{(4/3)}))^3 \approx 216$ person-years

- Interpretation:
- • Shorter schedules increase cost dramatically
- • Useful for long-term planning and estimation